MMET 410 – Lab 3

Fall 2026
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**Set up RSLogix 5000 software modules
**Have the worksheet pulled out on the side!

Lab 3A

1. Draw 2 diagrams and download them to machine.

1. Answer the following questions Q1 and Q2 for each task.

1. Only need to include the question and answer in your report. (Don't need the sequence of actions, tag table, and diagram)

Lab report format - 3B

- a. Problem Statements – Task Number
- b. Sequence of operations . Cover all possible branches
    1. LS2 is closed/on. Push Start button, then …..
    2. LS1 is closed/on. then ….
    3. LS2 is closed/on. then …
    4. …
    5. Push stop button, then ….

Lab report format - 3B

- c. Input and output devices and terminal address assignments

  Tag name and corresponding address

  For example:

Start	Local.8.I.Data.0
Stop	
LS1	

- d. Ladder logic program
  - Screenshot
  - Focus on main part.

Lab 3B-Task 1

1. Start bit =1 -> System ON
   **Stop bit = 0

1. System ON -> LS2 bit = 1 -> Conveyor FORWARD (CR1 ON)
   **LS1 bit = 0

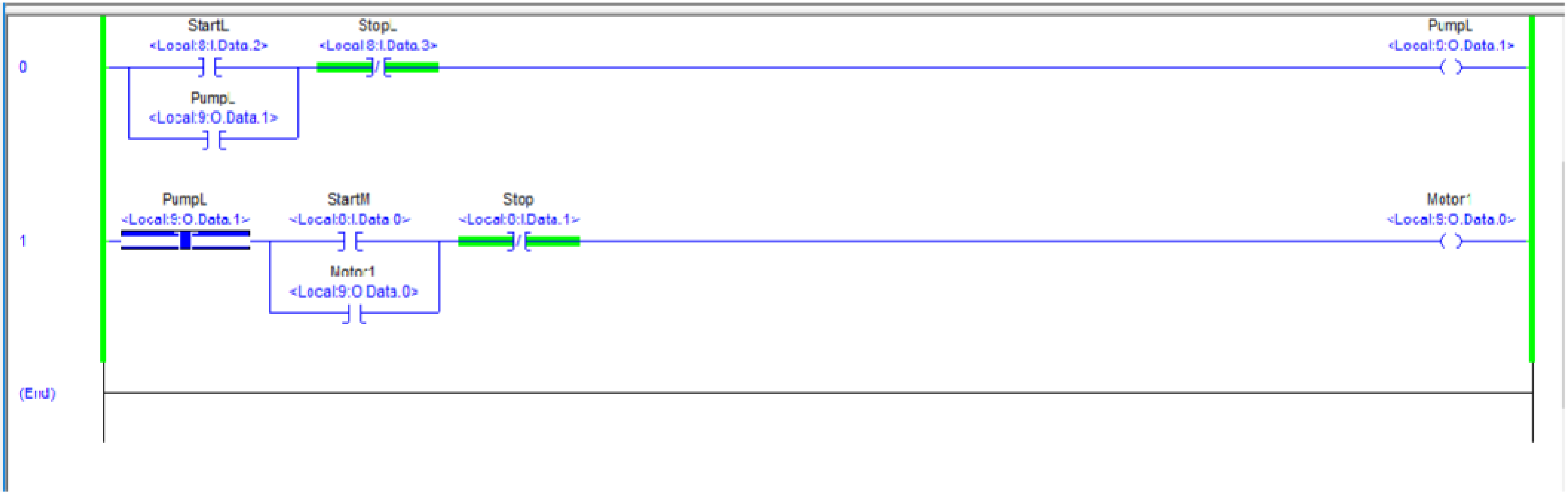
1. System ON -> LS1 bit = 1 -> Conveyor BACKWARD (CR2 ON)
   **LS2 bit = 0

1. Stop bit = 1 -> System OFF

- 3 rungs: **rung 0 is for system / rung 1 is for CR1 / rung 2 is for CR2**
- 4 inputs: Start / Stop / LS1 / LS2
- 3 outputs: System / CR1 / CR2

Lab 3B-Task 1

- **Spring return !!!**
- **Conditional activation !!!**
- **Same as what you did in task 4 lab 2**



Lab 3B-Task 1

Tags and addresses
IN-0 means local.8.I.Data.0 **start** is tag name
OUT-0 means local.9.O.Data.0 **system** is tag name

- For input, use **IN-0** as start, **IN-1** as stop, **IN-2** as LS1, **IN-3** as LS2
- For output, use **OUT-0** as system, **OUT-1** as CR1, **OUT-2** as CR2

Lab 3B – Task 2: Input Devices

3 pushbuttons + 2 toggle switches

Address	Tag	Hardware type	Notes
IN-0	DOOR_UP	Pushbutton (momentary, NO, spring return)	Bit is 1 only while pressed — held on by the MOTOR_UP seal-in
IN-1	DOOR_DOWN	Pushbutton	Same as above — held on by the MOTOR_DN seal-in
IN-2	STOP_DOOR	Pushbutton	Programmed as XIO — the rung is true only when it is NOT pressed
IN-4	UP_LIMIT	Toggle switch (maintained)	Flip on by hand to simulate the door reaching the top
IN-5	DOWN_LIMIT	Toggle switch (maintained)	Flip on by hand to simulate the door reaching the bottom

Why IN-3 is skipped: IN-3 is still a pushbutton, but a limit sensor has to hold its state — the DOOR_OPEN light must stay lit while the door sits at the top. So the two sensors move up to IN-4 and IN-5, the first maintained toggle switches on the trainer.

Lab 3B-Task 2

- 5 inputs: 3 Buttons: use IN-0 as Door_UP, IN-1 as Door_DN, IN-2 as Stop
  2 Switches to represent sensors: IN-4 as UP_limit, IN-5 as DN_limit
- 7 outputs:

  OUT-0 as AJAR, OUT-1 as Door_OPEN, OUT-2 as Door_CLOSED

  OUT-3 as Motor_UP, OUT-4 as Motor_DN,

  OUT-5 as Motor_UPWARD, OUT-6 as Motor DOWNWARD

- **5 rungs, first status lights three rungs are for(AJAR/OPEN/CLOSED)**, **the other two rungs are for Motor UPWARD/DOWNWARD actions**

- HINT: 3 examine off instructions for each of the FORWARD/DOWNWARD action

Lab 3B-Task 2

1. Door_UP bit = 1 -> Motor_UP True
   -> Motor_UPWARD True
   *** & **Stop bit=0** & **UP_limit bit =0** & **Motor_DOWNWARD bit =0**

1. Door_DN bit =1 -> Motor_DN True
   -> Motor_DOWNWARD True
   *** & **Stop bit=0** & **DN_limit bit=0** & **Motor_UPWARD bit =0**

1. Stop bit=1 -> System OFF

Before the motor activation, Indication lights first!

1. UP_limit Bit = 0 & DN_limit Bit = 0 -> AJAR True
2. UP_limit Bit = 1 -> Door_OPEN True
3. DN_limit Bit = 1 -> Door_CLOSED True

Lab 3B-Task 1

- **Same as what you did in task4 lab2, the only difference is that you need to add another rung**

Tags and addresses

IN-0 means local.8.I.Data.0 **start** is tag name
OUT-0 means local.9.O.Data.0 **system** is tag name

- For input, use **IN-0** as start, **IN-1** as stop, **IN-2** as LS1, **IN-3** as LS2
- For output, use **OUT-0** as system, **OUT-1** as CR1, **OUT-2** as CR2
- 3 rungs, 4 inputs, 3 outputs
- **rung 0 is for system**
- **rung 1 is for CR1**
- **rung 2 is for CR2**

Lab 3B-Task 2

5 inputs: 3 Buttons: use IN-0 as door up, IN-1 as door down, IN-2 as stop door

2 Switches to represent sensors: IN-4 as up limit, IN-5 as down limit

7 outputs: use OUT-0 as door ajar, OUT-1 as door open, OUT-2 as door closed, OUT-3 as motor up, OUT-4 as motor down, OUT-5 as motor upward, OUT-6 as motor downward

- Push door up button once will make the motor keep moving upward, and motor up also turns on.
- Push door down button once will make the motor keep moving downward, and motor down also turns on.
- Any of “door stop”, “up limit”, “motor down” can stop the up action
- Any of “door stop”, “down limit”, “motor up” can stop the down action
- (3 examine off instructions for each of the up/down action)
- **5 rungs, first three rungs are for status lights (up/down/ajar), the other two rungs are for door up/down actions**